Business Question and Visualization Report

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Team ID	EN25CS3011097
Project Name	Global Olympic Sports Performance Analysis

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

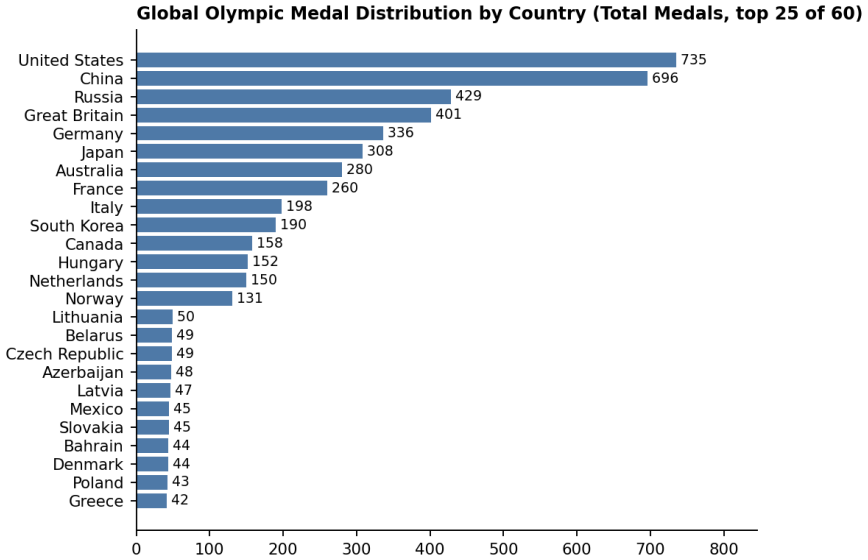
The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake. Well-framed business questions help identify the required metrics, determine the appropriate Snowflake tables or analytical views, select relevant data, and create visualizations that provide useful business insights. The data comes from **OLYMPIC_DB.SPORTS.OLYMPIC_EVENTS** (8,988 rows × 56 columns, Olympic data 1992–2026).

1. Which countries win the most Olympic medals overall?

Code

```
SELECT country, SUM(total_medals) AS total_medals
FROM OLYMPIC_DB.SPORTS.OLYMPIC_EVENTS
WHERE dataset_type = 'medal_results'
GROUP BY country
ORDER BY total_medals DESC;
```

Screenshot of visualisation



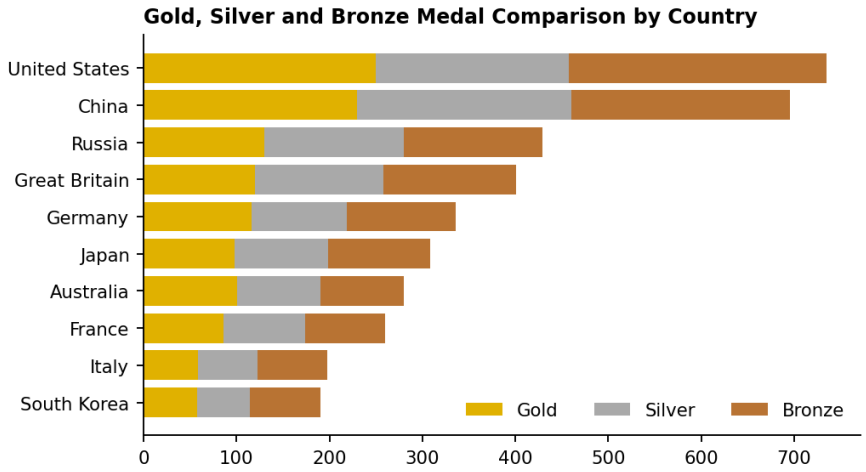
Insight: The United States (735) and China (696) lead by a wide margin, followed by Russia (429), Great Britain (401) and Germany (336).

2. How are gold, silver and bronze medals split among the leading countries?

Code

```
SELECT country, SUM(gold) AS gold, SUM(silver) AS silver, SUM(bronze) AS bronze
FROM OLYMPIC_DB.SPORTS.OLYMPIC_EVENTS
WHERE dataset_type = 'medal_results'
GROUP BY country
ORDER BY SUM(total_medals) DESC
LIMIT 10;
```

Screenshot of visualisation



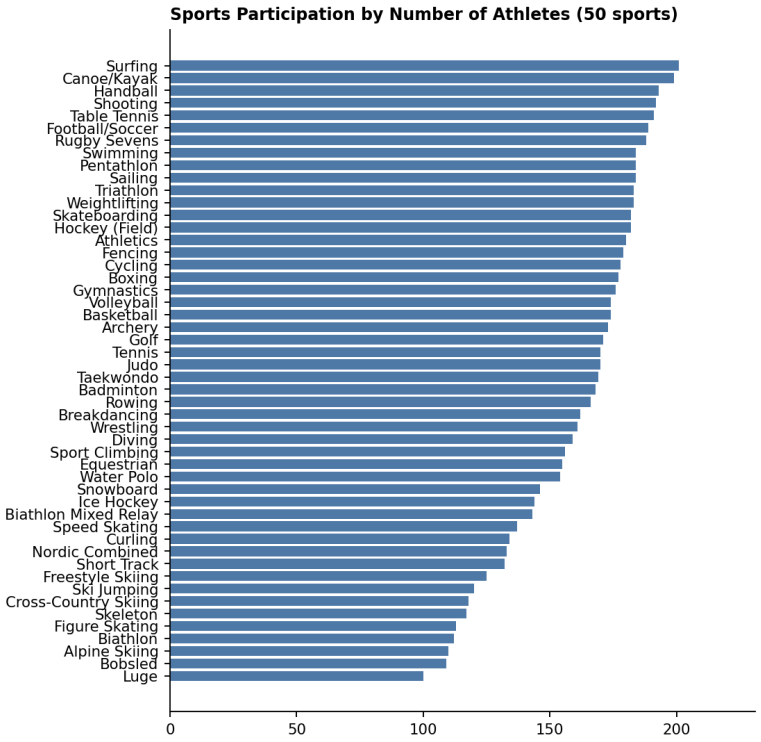
Insight: The gold race is close: United States 250 gold and China 230 gold, with silver and bronze adding up to 735 and 696 medals.

3. How many athletes take part in each of the 50 sports?

Code

```
SELECT sport, COUNT(DISTINCT athlete_id) AS athletes
FROM OLYMPIC_DB.SPORTS.OLYMPIC_EVENTS
WHERE dataset_type = 'athletes'
GROUP BY sport
ORDER BY athletes DESC;
```

Screenshot of visualisation



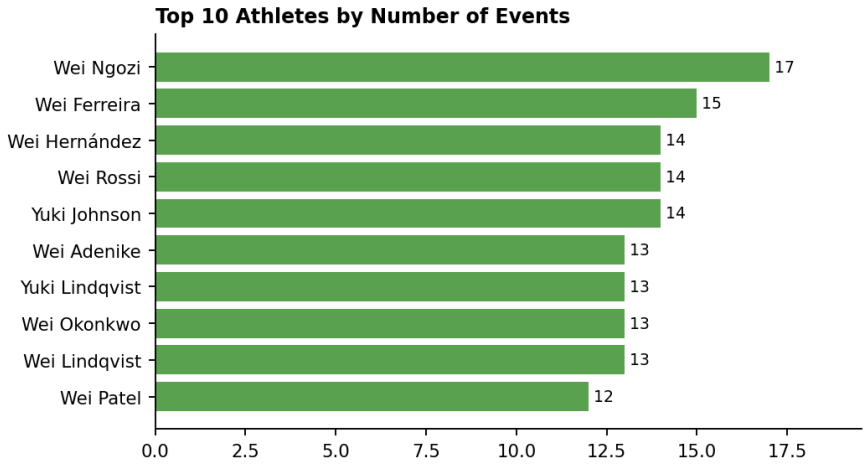
Insight: Participation is evenly spread, from 201 athletes in Surfing down to 100 in Luge.

4. Which athletes appear in the most events?

Code

```
SELECT name, COUNT(DISTINCT athlete_id) AS events
FROM OLYMPIC_DB.SPORTS.OLYMPIC_EVENTS
WHERE dataset_type = 'athletes'
GROUP BY name
ORDER BY events DESC, name
LIMIT 10;
```

Screenshot of visualisation



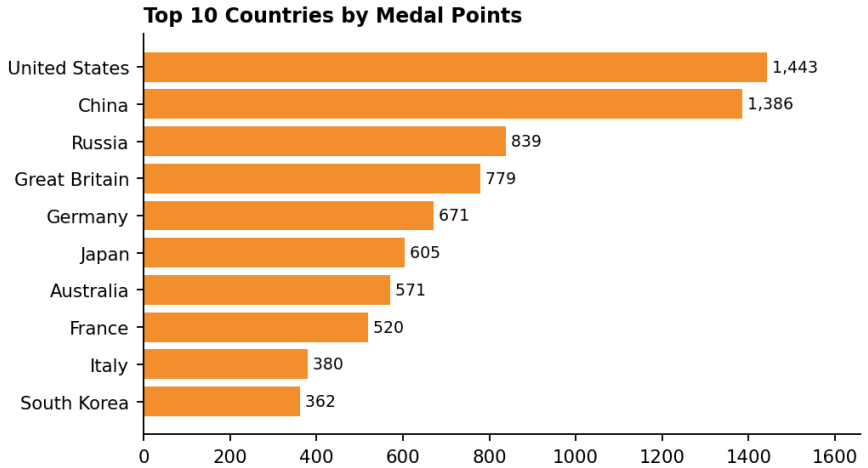
Insight: Wei Ngozi has the most records (17), followed by Wei Ferreira (15) and Wei Hernández, Wei Rossi and Yuki Johnson (14 each).

5. Which countries earn the most medal points across all Games?

Code

```
SELECT country, total_medal_points
FROM OLYMPIC_DB.SPORTS.OLYMPIC_EVENTS
WHERE dataset_type = 'alltime_country_rankings'
ORDER BY total_medal_points DESC
LIMIT 10;
```

Screenshot of visualisation



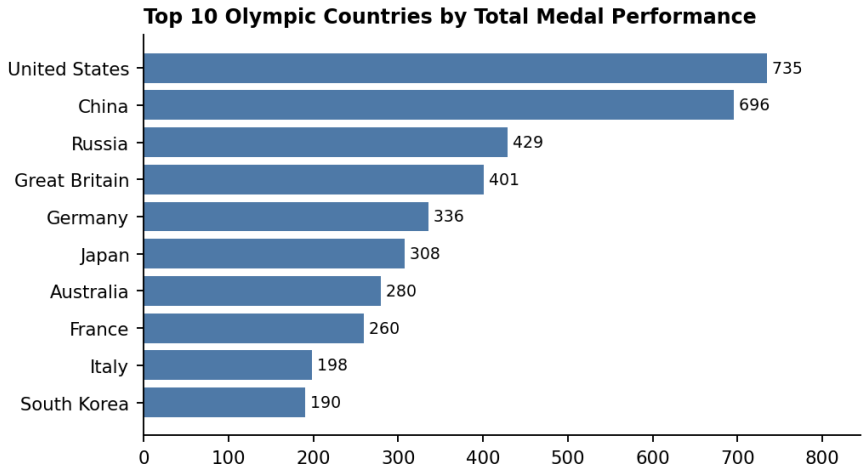
Insight: United States leads with 1,443 medal points, ahead of China (1,386); Russia follows with 839.

6. Which 10 countries have the best total medal performance?

Code

```
SELECT country, SUM(total_medals) AS total_medals
FROM OLYMPIC_DB.SPORTS.OLYMPIC_EVENTS
WHERE dataset_type = 'medal_results'
GROUP BY country
ORDER BY total_medals DESC
LIMIT 10;
```

Screenshot of visualisation



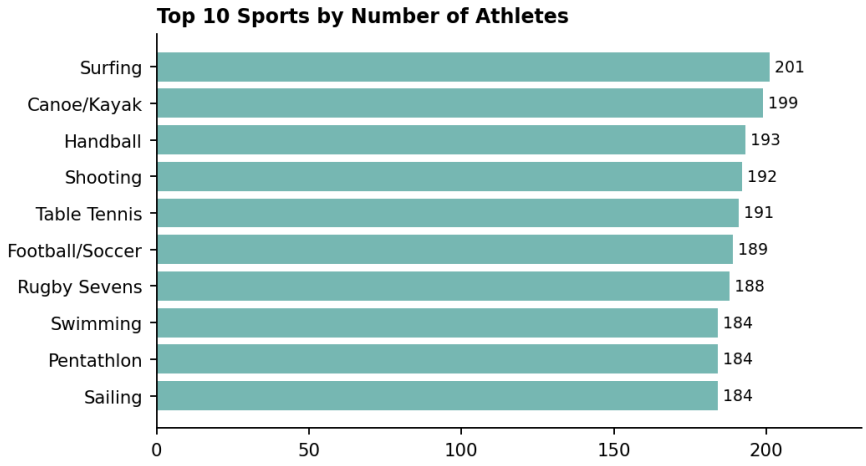
Insight: There is a large gap between the top two countries and the next group (Russia 429, Great Britain 401, Germany 336).

7. Which 10 sports attract the most athletes?

Code

```
SELECT sport, COUNT(DISTINCT athlete_id) AS athletes
FROM OLYMPIC_DB.SPORTS.OLYMPIC_EVENTS
WHERE dataset_type = 'athletes'
GROUP BY sport
ORDER BY athletes DESC
LIMIT 10;
```

Screenshot of visualisation



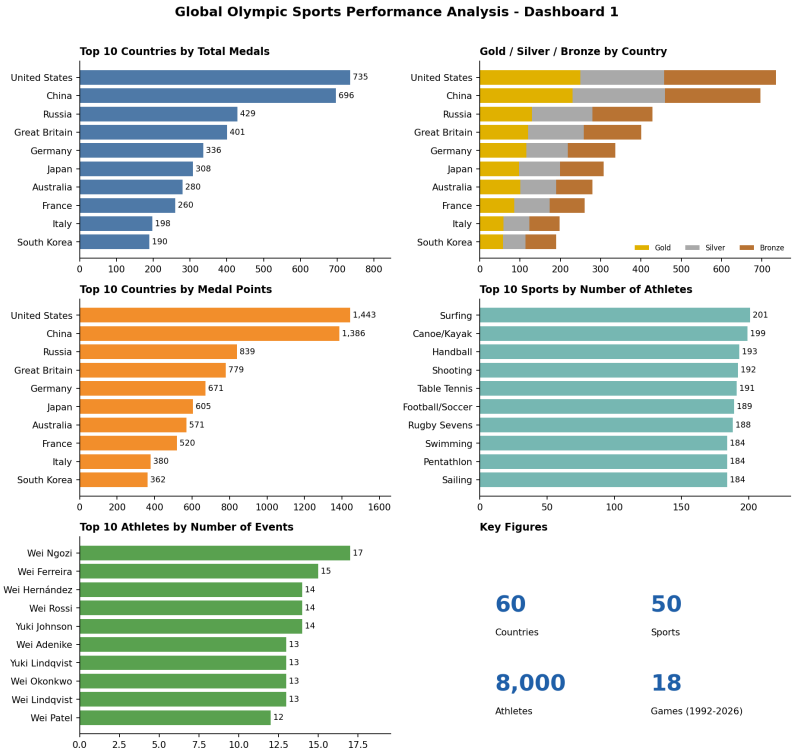
Insight: Surfing (201), Canoe/Kayak (199) and Handball (193) have the most athletes; the Top 10 sit between 184 and 201.

8. What is the overall Olympic performance picture across countries, sports and athletes?

Code

```
SELECT
 (SELECT COUNT(DISTINCT country)    FROM OLYMPIC_EVENTS WHERE dataset_type='medal_results') AS countries,
 (SELECT COUNT(DISTINCT sport)      FROM OLYMPIC_EVENTS WHERE dataset_type='athletes')      AS sports,
 (SELECT COUNT(DISTINCT athlete_id) FROM OLYMPIC_EVENTS WHERE dataset_type='athletes')      AS athletes,
 (SELECT COUNT(*)                   FROM OLYMPIC_EVENTS WHERE dataset_type='games_summary') AS games;
```

Screenshot of visualisation



Insight: Key figures: 60 countries, 50 sports, 8,000 athletes and 18 Games (1992–2026, 9 Summer and 9 Winter).